

# Manual

## Real-time Process Data Monitoring PDV-RMO 8.3

Version 1.0.23049

Last changed on: 02.09.2020

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# 1 Real-Time Process Data Monitoring

## Overview

### Purpose

The product PDV-RMO includes visualization functions for the MOC client. The data visualized is the process data that has been distributed via MQTT broker.

The logical channels and the collection requests that are actively performed at the machines specify which data can be displayed.

To visualize data, predefined display instruments and trend graphs are available. You can use special layouts for the visualization.

### Integration

This product requires a machine interface for data collection in the Process Communication Controller (PCC), the network infrastructure and the basic PDV-PDM package including collection rules.

### Features

- This product provides a permanent display of process data of one or several machines or systems.
- This product provides an online display of measured values including predefined gauge charts for a machine.
- This product provides an online display of measured values as trend line for a machine. The last 50 measured values of a characteristic recorded since the start of the trend are visualized as continuous trend chart.

## 2 Process Visualization

### Overview

|                        |                                                               |
|------------------------|---------------------------------------------------------------|
| Menu                   | Quality management → Process analysis → Process visualization |
| Transaction code       | vispd                                                         |
| Function authorization | visupd                                                        |

### Purpose

Using the data that is collected according to defined collection rules, you can visualize defined characteristics online. In the network, an MQTT Broker is used to provide the characteristics that are selected for visualization.

To display the values, select one of the predefined templates as layout. You can also integrate individual display layouts.

### Integration

The process data provided by the MQTT Broker is visualized using the display functions. The visualization is based on the specifications made for the data collection.

### Requirements

If the option *Visualize* is enabled in the collection rule of the process characteristic, then this process characteristic is provided by the visualization component.

An MQTT Broker provides the process data that must be visualized and transfers them to the MOC visualization component. A continuous Ethernet connection between MOC and MQTT Broker must be provided.

### Selection criteria

The application provides the following selection criteria:

#### Workplace

Select the workplace using the selection field *Workplace*. The process parameters of this workplace are then visualized. You can select a machine.

## Layout

The dropdown selection list of the layout includes the available display layouts. Select a layout from the dropdown selection list.

You can select one of the HYDRA standard layouts or a custom layout, if required.

## Detail applications

In the *Visualization* panel, the process parameters of the selected workplace that are specified for visualization are displayed. The selected layout specifies the display of the elements.

The scaling of the display element depends on the recorded process parameter value and its specified limits (target value, process action limit and tolerance limit).

### Trend chart:

The measured values of a characteristic recorded since the start of the trend are visualized as continuous trend chart. The last 50 measured values recorded can be displayed at most.